SECTIONS SUPPORTED ON A CLOSED SUBSET WITH RETROCOMPACT COMPLEMENT FORM A QUASI-COHERENT SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the finitely generated ideals provided by retrocompactness, the translation of the support condition into annihilation by a power of the ideal through the support lemma, and the identification of the sheaf as a kernel of a map to a quasi-coherent pushforward.

Editorial note. The label lemma-sections-supported-on-closed-subset names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-sections-supported-on-closed-subset –start 4051 –end 4108 –local-id 28038 returned RESULT=OK (L28038 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Finiteness of the ideal defining a closed subset in an affine open. Algebra, the lemma that a closed subset of an affine scheme with retrocompact complement is defined by a finitely generated ideal. Supplies the finitely generated ideal I with Z cap U = V(I).
- Description of the support of a section and annihilation by the ideal. Modules, the definition of the support of a section, and Algebra, the lemma relating the vanishing of a localization to the containment of the support in a closed subset. Supplies the equivalence between the support of x being contained in V(I) and the annihilation of x by some power of I.
- Quasi-coherence of the pushforward along a quasi-compact open immersion. Schemes, the lemma that the pushforward along a quasi-compact and separated morphism of a quasi-coherent module is quasi-coherent. Supplies the quasi-coherence of the pushforward of the restriction to the complement, so that the kernel of the displayed map is quasi-coherent.

Source reference. Original lemma 00F6: sources/stacks-chow/tags.txt, source lines 560–560. Prerequisite pointer invoked at proof line(s) 4075; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07Z5: sources/stacks-chow/tags.txt, source lines 10135–10135. Prerequisite pointer invoked at proof line(s) 4085; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 01AT: sources/stacks-chow/tags.txt, source lines 1702–1702. Prerequisite pointer invoked at proof line(s) 4083; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Prerequisite pointer invoked at proof line(s) 4103; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 01LA: sources/stacks-chow/tags.txt, source lines 2079–2079. Prerequisite pointer invoked at proof line(s) 4107; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Let $Z \subset X$ be a closed subset. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Consider the sheaf of $\mathcal{O}_X$-modules $\mathcal{F}'$ which associates to every open $U \subset X$*

$$\mathcal{F}'(U) = \{s \in \mathcal{F}(U) \mid \textit{the support of } s \textit{ is contained in } Z \cap U\}$$

If $X \setminus Z$ is a retrocompact open of X, then

1. *for an affine open $U \subset X$ there exist a finitely generated ideal $I \subset \mathcal{O}_X(U)$ such that $Z \cap U = V(I)$,*
2. *for U and I as in (1) we have $\mathcal{F}'(U) = \{x \in \mathcal{F}(U) \mid I^n x = 0 \text{ for some } n\}$,*
3. *$\mathcal{F}'$ is a quasi-coherent sheaf of $\mathcal{O}_X$-modules.*

Proof. Part (1) is Algebra, Lemma 00F6. Let $U = \mathrm{Spec}(A)$ and I be as in (1). Then $\mathcal{F}|_U$ is the quasi-coherent sheaf associated to some A-module M. We have

$$\mathcal{F}'(U) = \{x \in M \mid x = 0 \text{ in } M_{\mathfrak{p}} \text{ for all } \mathfrak{p} \notin Z\}.$$

by Modules, Definition 01AT. Thus $x \in \mathcal{F}'(U)$ if and only if $V(\mathrm{Ann}(x)) \subset V(I)$, see Algebra, Lemma 07Z5. Since I is finitely generated this is equivalent to $I^n x = 0$ for some n. This proves (2).

Proof of (3). Observe that given $U \subset X$ open there is an exact sequence

$$0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}(U \setminus Z)$$

If we denote $j : X \setminus Z \to X$ the inclusion morphism, then we observe that $\mathcal{F}(U \setminus Z)$ is the sections of the module $j_*(\mathcal{F}|_{X \setminus Z})$ over U. Thus we have an exact sequence

$$0 \to \mathcal{F}' \to \mathcal{F} \to j_*(\mathcal{F}|_{X \setminus Z})$$

The restriction $\mathcal{F}|_{X \setminus Z}$ is quasi-coherent. Hence $j_*(\mathcal{F}|_{X \setminus Z})$ is quasi-coherent by Schemes, Lemma 01LC and our assumption that j is quasi-compact (any open immersion is separated). Hence $\mathcal{F}'$ is quasi-coherent as a kernel of a map of quasi-coherent modules, see Schemes, Section 01LA. □